CENG381 Stochastic Processes -- Mixed Exam

Mixed Exam 1 -- Markov Chains | Biased Random Walk | Poisson Process

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Duration: 90 minutes. Total: 100 points. Show all working.

Notation: P(i,j) = one-step transition probability from state i to j. pi(i) = stationary probability of state i. S(n) = position after n steps. N(t) = number of Poisson arrivals by time t.

Q1 (30 points). A Markov chain has three states {A, B, C} with transition matrix:

$$
\begin{array}{c c}
 & \begin{array}{ccc} A & B & C \end{array} \\
\begin{array}{c} A \\ B \\ C \end{array} & \left[\begin{array}{ccc} 0.5 & 0.3 & 0.2 \\ 0.1 & 0.6 & 0.3 \\ 0.4 & 0.1 & 0.5 \end{array} \right]
\end{array}
$$

a) Is this chain irreducible? Explain by showing that every state can reach every other state in one or more steps.

b) Find the stationary distribution pi = (pi(A), pi(B), pi(C)) by solving the system pi * P = pi together with pi(A) + pi(B) + pi(C) = 1. Set up the three linear equations explicitly before solving.

c) Starting in state A, what is the expected number of steps to return to A for the first time? Use the formula: E[return time to A] = 1 / pi(A).

Q2 (35 points). A particle performs a biased random walk on {0, 1, 2, 3, 4, 5}. At each interior state k (with 1 <= k <= 4) the particle moves right with probability p = 2/3 and left with probability q = 1/3. States 0 and 5 are absorbing (the particle stops on arrival).

a) Write the hitting probability recurrence for h(k) = P(reach 5 before 0 | start at k):

h(k) = p * h(k+1) + q * h(k-1)   for k = 1, 2, 3, 4

with boundary conditions h(0) = 0 and h(5) = 1.

The general solution is h(k) = A + B * r^k where r = q/p. Identify r and use the boundary conditions to find A and B.

b) Compute h(2) = P(reach 5 before 0 | start at k = 2). Compare with the symmetric walk result h_sym(2) = 2/5.

c) For the symmetric walk (p = q = 1/2, starting at k = 2, N = 5): use the Optional Stopping Theorem with martingale S(n) to confirm h_sym(2) = 2/5, and with martingale S(n)^2 - n to find E[T | start k = 2] where T is the absorption time.

Q3 (35 points). Customers arrive at a coffee shop according to a Poisson process

with rate lambda = 4 per hour. Each customer independently orders a latte with probability p = 0.25.

a) State the Thinning Property: what is the rate lambda_L of the latte-ordering sub-process? What distribution does N_L(t) (number of latte orders by time t) follow?

b) Find the probability that NO latte is ordered in the first 30 minutes (t = 0.5 hours). Use the Poisson PMF with n = 0:

P(N_L(0.5) = 0) = e^(-lambda_L * 0.5)

c) Given that exactly 8 customers arrive in the first hour, what is the conditional distribution of the number of latte orders? State the distribution by name and give its parameters. Find the conditional expected number of latte orders.